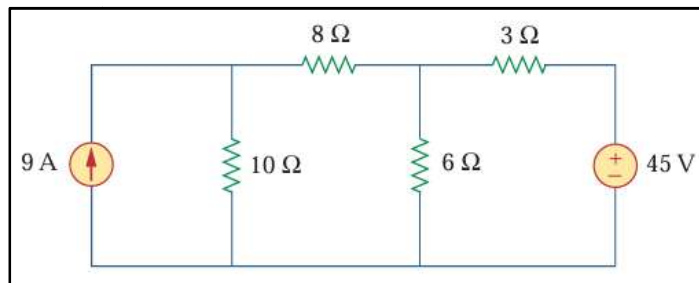


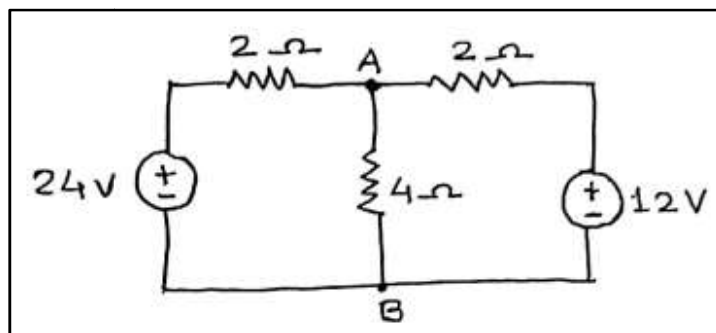
EE10510: Fundamentals of Electrical and Electronics Engineering
Assignment Sheet-I
Based on Unit I (DC Circuits)

Max Marks: 5
Due Date: 07.11.2025

1. Write the statement of – Ohm's law, KCL, KVL, Superposition Theorem, Thevenin's Theorem, Norton's Theorem and Maximum power transfer Theorem.
2. Compare the resistor, inductor and capacitor on the basis of their circuit symbol, electrical properties and uses.
3. Use source transformation to determine the current and power in the 8Ω resistor.



4. Determine the current through 4Ω resistor in the network as shown in figure, by (i) Nodal analysis, (ii) Mesh analysis, (iii) Superposition theorem, (iv) Thevenin's theorem, (v) Norton's theorem. Also apply maximum power transfer theorem to find the value of $P_{\max}$.



5. Find R_{ab} and i using star-delta transformation.

